SRIVARDHAN S

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Overview

Robotics and Artificial Intelligence undergraduate with hands-on experience in autonomous UAV development, Model Predictive Control, and machine learning. Interested in robotics research, autonomous systems, control engineering, and intelligent flight systems.

Education

Amrita Vishwa Vidyapeetham Chennai, India
Bachelor of Technology – Robotics and Artificial Intelligence *2024 – 2028*
Relevant Coursework: Control Systems, Robot Kinematics and Dynamics, Machine Learning, Sensor Systems, Embedded Systems, Design and Analysis.

Experience

Brakes India Pvt. Ltd. – AMTIX Division, Chennai *2024*
Engineering Intern

- Studied sensor-based automated workflows and closed-loop feedback mechanisms relevant to precision control and robotic actuation systems.
- Observed quality control standards, industrial automation, and safety protocols in high-precision manufacturing environments.

Projects

Battery-Aware Model Predictive Control for Autonomous UAVs *2026*
Python | CVXPY | OSQP | NumPy | MPC | Optimization

- Developed a dual-loop Model Predictive Control (MPC) framework for autonomous quadrotor trajectory planning using a discrete-time state-space model.
- Formulated constrained quadratic optimization problems using CVXPY and OSQP for trajectory tracking and obstacle avoidance.
- Designed a jerk-aware cost function and battery State-of-Charge (SoC) model to reduce aggressive control actions and improve energy efficiency.
- Evaluated five UAV mission profiles, achieving up to 25.3% lower control energy, 57.6% lower jerk, and 22.7% lower battery drain compared with baseline MPC.
- Conducted prediction horizon ablation and ±30% parameter sensitivity analysis to evaluate controller robustness and computational trade-offs.

Autonomous UAV Development and Waypoint Navigation *2026*
CUAV X7+ | ArduPilot | Mission Planner | Lua Scripting

- Designed and developed an autonomous drone using the CUAV X7+ flight controller, configured and monitored through Mission Planner.
- Implemented GPS-based waypoint navigation and custom Lua scripts for autonomous flight operations.

Anomaly Detection in UAV Systems *2026*
Python | scikit-learn | NumPy | pandas | Sensor Data Processing

- Developed a machine learning pipeline for anomaly detection using UAV flight and sensor data.
- Applied feature engineering, data preprocessing, classification, and model evaluation for reliable UAV system monitoring.

Publications and Intellectual Property

Registered Design – Expandable Wall Hanger *2026*
Intellectual Property Office, Government of India

- Co-registered designer under Design Registration No. 512390-001, Class 06-08, under the Designs Act, 2000.
- Registration dated 06 August 2026; certificate issued on 23 September 2026.

Technical Skills

Programming Languages: Python, C (Basic), C++ (Basic), MySQL.

Controls and Optimization: Model Predictive Control (MPC), CVXPY, OSQP, Trajectory Planning, Quadratic Optimization.

Robotics and UAV Tools: CUAV X7+, Pixhawk, Mission Planner, ArduPilot, Lua Scripting, ANSYS Mechanical, Fusion 360, ROS2, Gazebo.

Embedded and IoT Systems: Hardware-Software Integration, Sensor Interfacing, Embedded Systems, IoT Platforms.

Leadership and Volunteering

Tantrotsav '25 – Paper Presentation Event Co-host *2025*
Amrita Vishwa Vidyapeetham, Chennai

- Coordinated a technical paper presentation event involving 40+ participants across engineering domains, including robotics and control systems.
- Reviewed technical abstracts and moderated Q&A sessions between presenters and faculty judges.

Additional Information

Languages: English, Tamil.